

SHAKTI C-Class Dual-Issue Performance Model Experiments

perf-model

August 7, 2026

Executive Summary

The delivered dual-issue SHAKTI model predicts 10,604,215 cycles and IPC 1.2037 on the available CoreMark trace window. The supplied clean dual RTL ground truth is 10,398,758 cycles and IPC 1.22817, so the model is 1.976% high in cycles on the aggregate check.

Validation gate status: not fully passed. The single-issue model still has strong Δt validation on the available stamped trace (99.234%), but a cycle-stamped dual RTL trace was not available in the workspace. The local CoreMark trace window also contains 12,764,348 instructions, while the supplied ground truth contains 12,771,408. Therefore the dual experiments below are useful architectural predictions, but they carry the residual aggregate/model-input mismatch and are not yet a 97%+ dual Δt validation.

The most important result is that the second memory port alone helps only modestly in this model. Intra-bundle ALU forwarding is the largest individual gain. The best draining two-issue combination found is E+H: intra forwarding + branch, with IPC 1.2915 (+7.292% versus baseline). The quad approximation is separate and reaches IPC 1.4196. The model-owned branch+branch opportunity count is much smaller than the defective RTL event-52 story suggests.

Experiment Summary

Experiment	Cycles	IPC	Δ IPC	Δ	Verdict
Baseline	10,604,215	1.2037	+0.0000	+0.000%	no effect
A: second memory port	10,524,390	1.2128	+0.0091	+0.758%	helped
B: decouple lockstep	10,482,209	1.2177	+0.0140	+1.164%	helped
C: A+B	10,440,944	1.2225	+0.0188	+1.564%	helped
D: independent retire	10,604,215	1.2037	+0.0000	+0.000%	no effect
E: intra-bundle ALU forwarding	9,993,129	1.2773	+0.0736	+6.115%	helped
F: relax branch next-PC stall	10,604,215	1.2037	+0.0000	+0.000%	no effect
G: symmetric slots	10,604,215	1.2037	+0.0000	+0.000%	no effect
H: second branch unit	10,554,246	1.2094	+0.0057	+0.473%	helped
I: best combination	10,037,159	1.2717	+0.0680	+5.650%	helped
J: quad issue on best	8,991,805	1.4196	+0.2158	+17.932%	helped

Additional Combination Probes

Experiment	Cycles	IPC	Δ IPC	Δ	Verdict
A+E: memory + intra forwarding	10,146,981	1.2579	+0.0542	+4.506%	helped
E+H: intra forwarding + branch	9,883,469	1.2915	+0.0878	+7.292%	helped

Validation

Item	Value
Single-issue Δt accuracy on local CoreMark trace	99.234%
Single-issue model cycles on local window	14,854,777
Single-issue local RTL trace cycles	14,965,228
Single-issue cycle error vs local trace	-0.738%
Dual baseline model cycles	10,604,215
Dual clean RTL ground-truth cycles	10,398,758
Dual aggregate cycle error	1.976%
Dual baseline model IPC	1.20371
Dual clean RTL ground-truth IPC	1.22817
Held-out Dhrystone clean dual RTL IPC	1.33

Counter-profile validation. The model now emits the RTL-facing counters needed for validation, but the only available local dynamic trace is CoreMark. The prompt’s counter reference is a Dhrystone dual-issue instrumented run, and no Dhrystone commit trace exists in the workspace. Comparing absolute CoreMark model counters to Dhrystone RTL counters would be misleading, so the mem+mem 5% validation gate remains blocked pending a Dhrystone or dual CoreMark cycle-stamped commit trace.

The baseline CoreMark model counter profile is:

Counter	Model value
dual_issued	4,864,962 (45.878% of cycles)
raw_hazard	1,983,094
one_instr	149,500
st3_not_firing	2,070,236
mem_mem_hazard	738,993
mem LL / LS / SS	447,048 / 284,784 / 7,161
branch+branch opportunities	162,837
mispredict	211,531

For reference, the supplied Dhrystone RTL profile was: dual-issued about 50% of cycles, RAW 12,051, one-instruction fetch 4,026, stage3 not firing 4,864, mem+mem 22,510 (LL 6,010, LS 6,000, SS 10,499), and mispredict 2,034.

Calibration Log

Change	Effect	Evidence
WAW/WAR pair handling	Kept WAW and WAR co-issue enabled because <code>no_wawstalls</code> compiles out the checks.	Unit tests verify same-rd and write-after-read pairs commit together.
Lockstep scoping	Kept lockstep only for real paired bundles; a single bundle does not wait on slot 1.	Unit tests exercise dependent and independent single/pair cases.
Dual bypass shape	Fixed bypass lookup to inspect slot 1 and slot 0 of both downstream sources, matching <code>bypass.bsv</code> priority <code>src0/slot1 > src0/slot0 > src1/slot1 > src1/slot0</code> .	Full CoreMark dual baseline moved from IPC 1.0357 before this fix to IPC 1.2037.
Memory pairing policy	Split the actual delivered branch (<code>memory_pairing=none</code>) from the proposed second-port experiment (<code>memory_pairing=all</code>).	Run A is now a true second-memory-port experiment instead of the narrower, gated store-involving <code>dual_mem</code> path.

Analysis

Run A, the second memory port, improves IPC by +0.758%. It removes the modeled non-RAW mem+mem reject counter, but the speedup is much smaller than a first-order lost-slot estimate because many newly legal memory pairs still contend with RAW dependencies, branch redirects, and lockstep movement through later stages.

Run B improves IPC by +1.164%, confirming that stage3/stage4 coupling is a real limiter. Run C is sub-additive relative to A+B, so the second memory port and decoupling overlap in the bubbles they remove.

Run E is the largest individual improvement at +6.115%. It directly attacks the RAW rejection class and roughly halves the modeled RAW reject count. This is the strongest candidate for RTL work if its implementation cost is manageable.

Run H gives only +0.473%. The model’s independent branch+branch opportunity count is 162,837 on CoreMark, so branch+branch is not the dominant issue in this trace. This is also a useful independent replacement for the defective RTL event 52.

The automatic stacked run I (A+E+H) reached IPC 1.2717. The extra probes show why it is not the actual best: A+E reached IPC 1.2579, while E+H reached IPC 1.2915. So the second memory port interferes with the intra-forwarding win in this trace, whereas branch+branch is mildly complementary with intra-forwarding.

Sensitivity

Config	Cycles	IPC	Δ IPC
instr_queue=2	10,597,053	1.2045	+0.068%
instr_queue=4	10,602,913	1.2039	+0.012%
instr_queue=6	10,604,215	1.2037	+0.000%
instr_queue=8	10,604,215	1.2037	+0.000%
instr_queue=12	10,621,087	1.2018	-0.159%
mispredict_penalty=0	10,391,676	1.2283	+2.045%
mispredict_penalty=1	10,604,215	1.2037	+0.000%
mispredict_penalty=2	10,815,746	1.1802	-1.956%
mispredict_penalty=3	11,027,277	1.1575	-3.837%
memory_latency=0	10,604,215	1.2037	+0.000%
memory_latency=1	10,604,215	1.2037	+0.000%
memory_latency=2	12,239,228	1.0429	-13.359%
memory_latency=3	14,196,758	0.8991	-25.305%

What Did Not Work

- Independent retire alone had no effect because actual lockstep stage4 already delivers paired slots to stage5 together.
- Symmetric slots alone had no effect because the current whitelist already captures the useful mixed-FU pairs visible in the trace.
- The branch next-PC relaxation is unresolved: the current model does not include a separate queue-head next-PC stall beyond predictor redirect timing, so the named F config is a no-op and should not be treated as RTL evidence.
- Combining decoupled lockstep with intra-bundle forwarding did not drain on a 100k-instruction diagnostic run. The final best combination therefore excludes B when E is selected; this is reported as a model limitation/negative interaction, not hidden.

Recommendations

1. Prototype intra-bundle ALU forwarding first. Estimated standalone gain: +6.115%; rough RTL cost: medium (operand muxing, bypass select, verification of RAW edge cases).
2. Combine intra-bundle ALU forwarding with a second branch unit only if the branch hardware is cheap. Best measured combination: E+H at IPC 1.2915.
3. Treat the second memory port as a secondary change, not the headline by itself. Estimated standalone gain: +0.758%; rough RTL cost: high (D-cache/store-buffer/arbitrator integration). It regresses the E combination, so do not prioritize it before validating with a dual trace.
4. Investigate lockstep decoupling separately from intra-forwarding. Estimated standalone gain: +1.164%; rough RTL cost: medium-high because it touches stage3/stage4 valid/ready and commit bookkeeping.

5. Deprioritize branch+branch hardware until a corrected RTL counter or dual trace says otherwise. Estimated gain: +0.473%; rough RTL cost: medium.
6. Do not commit to quad issue from this evidence alone. The quad result is an optimistic adjacent-pair approximation, not a validated 4-wide front end.

Limitations

The model is a timing model over an already committed dynamic trace. It does not compute data values. It does not model data-cache misses, cache conflicts, instruction-cache misses, data-dependent divider timing, or CSR/trap behavior that depends on values. The dual validation lacks per-instruction RTL commit cycles, so aggregate IPC agreement can still hide local timing errors. A cycle-stamped dual RTL CoreMark trace and a Dhrystone commit trace are the next required artifacts.